

13. | Implement your own ping program.

AIM: To implement our own ping program in Python that resolves a host name to its IP address and measures the response time of the host.

ping_program.py

```
import socket
```

```
import time
```

```
host = input("Enter host/IP address: ")
```

```
try:
```

```
    ip = socket.gethostbyname(host)
```

```
    print("Pinging", host, "[", ip, "]")
```

```
    for i in range(4):
```

```
        start = time.time()
```

```
        try:
```

```
            s = socket.create_connection((ip, 80), timeout=2)
```

```
            end = time.time()
```

```
            print("Reply from", ip, ": time =", round((end - start) * 1000, 2), "ms")
```

```
            s.close()
```

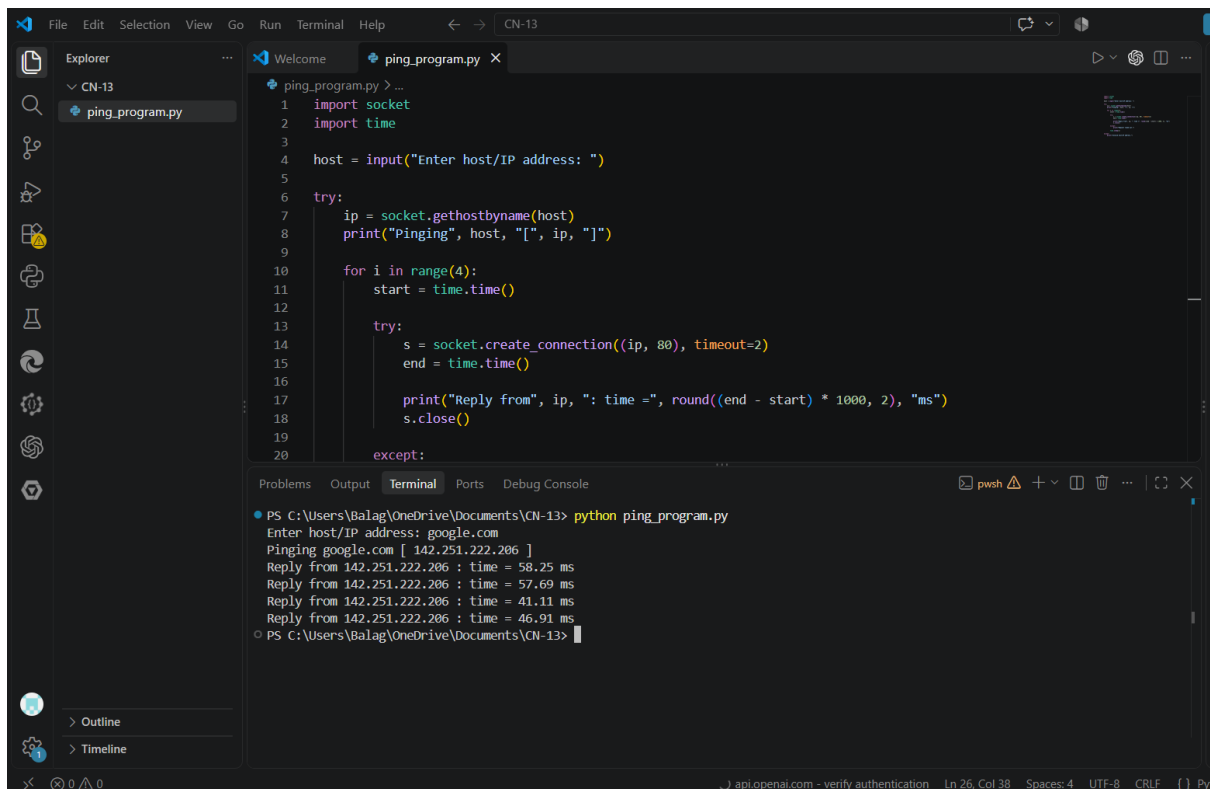
```
except:
```

```
print("Request timed out.")
```

```
time.sleep(1)
```

except:

```
print("Invalid host/IP address.")
```



The screenshot shows a Visual Studio Code editor window with a file named `ping_program.py` open. The code is a Python script that takes a host/IP address as input and pings it four times, displaying the response time in milliseconds. The terminal output shows the script being executed with `python ping_program.py` and the user entering `google.com`. The output shows four successful ping responses from IP `142.251.222.206` with response times of 58.25 ms, 57.69 ms, 41.11 ms, and 46.91 ms.

```
1 import socket
2 import time
3
4 host = input("Enter host/IP address: ")
5
6 try:
7     ip = socket.gethostbyname(host)
8     print("Pinging", host, "[", ip, "]")
9
10    for i in range(4):
11        start = time.time()
12
13        try:
14            s = socket.create_connection((ip, 80), timeout=2)
15            end = time.time()
16
17            print("Reply from", ip, ": time =", round((end - start) * 1000, 2), "ms")
18            s.close()
19
20 except:
```

```
PS C:\Users\Balag\OneDrive\Documents\CN-13> python ping_program.py
Enter host/IP address: google.com
Pinging google.com [ 142.251.222.206 ]
Reply from 142.251.222.206 : time = 58.25 ms
Reply from 142.251.222.206 : time = 57.69 ms
Reply from 142.251.222.206 : time = 41.11 ms
Reply from 142.251.222.206 : time = 46.91 ms
PS C:\Users\Balag\OneDrive\Documents\CN-13>
```

```
● PS C:\Users\Balag\OneDrive\Documents\CN-13> python ping_program.py
Enter host/IP address: google.com
Pinging google.com [ 142.251.222.206 ]
Reply from 142.251.222.206 : time = 58.25 ms
Reply from 142.251.222.206 : time = 57.69 ms
Reply from 142.251.222.206 : time = 41.11 ms
Reply from 142.251.222.206 : time = 46.91 ms
○ PS C:\Users\Balag\OneDrive\Documents\CN-13>
```